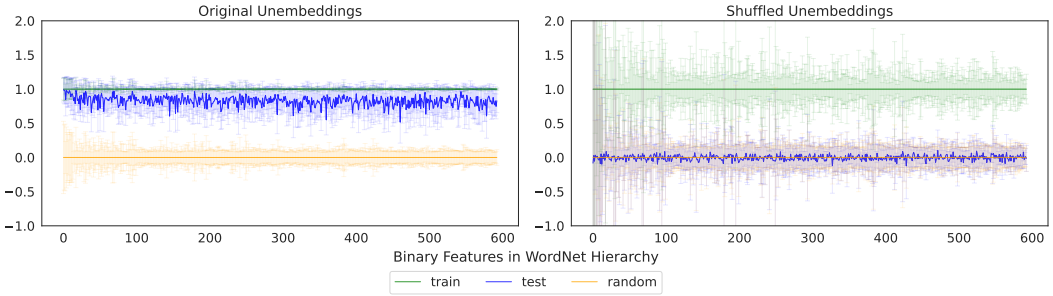


Figure 3: Vector representations exist for most binary features in the WordNet noun hierarchy. For each synset $w$ (indexed on the $x$-axis) we estimate the vector representation $\bar{\ell}_w$ using a train subset of the vocabulary $\mathcal{Y}(w)$. The plot shows the projections $(g(y)^\top \bar{\ell}_w)/\|\bar{\ell}_w\|_2^2$ of train (green), test (blue), and random (orange) words on estimated vector representations for each WordNet feature, using either the original (left) or shuffled (right) unembeddings. Our theory predicts that this value should be close to 1 when $y$ has the target feature, and close to 0 when it does not. The thick lines present the mean of the projections for each feature and the error bars indicate the standard deviation. As predicted, the projections of test words are near 1, and random words near 0 (left plot). Further, this structure does not hold when using the shuffled control without natural semantics (right plot).

**WordNet** We define a large collection of binary concepts using WordNet (Miller, 1995). WordNet organizes English words into a hierarchy of synsets, where each synset is a set of synonyms. The WordNet hierarchy is based on word hyponym relations, and reflects the semantic hierarchy of interest in this paper. We take each synset as an attribute $w$ and define $\mathcal{Y}(w)$ as the collection of all words belonging to any synset that is a descendant of $w$. For example, the synset `mammal.n.01` is a descendant of `animal.n.01`, so both $\mathcal{Y}$(`mammal.n.01`) and $\mathcal{Y}$(`animal.n.01`) contain the word "dog". We collect all noun and verb synsets, and augment the word collections by including plural forms of the nouns, multiple tenses of each verb, and capital and lower case versions of each word. We filter to include only those synsets with at least 50 words in the Gemma vocabulary. This leaves us with 593 noun and 364 verb synsets, each defining an attribute.

For space, we report results on the noun hierarchy here and defer the verb hierarchy to Appendix F.

**Estimation via Linear Discriminant Analysis** Now, we want to estimate the vector representation $\bar{\ell}_w$ for each attribute $w$. To do this, we make use of vocabulary sets $\mathcal{Y}(w)$. Following Theorem 4, the vector associated to the concept $w$ should have two properties. First, when the full vocabulary is projected onto this vector, the words in $\mathcal{Y}(w)$ should be well-separated from the rest of the vocabulary. Second, the projection of the unembedding vectors for $y \in \mathcal{Y}(w)$ should be approximately the same value. Equivalently, the variance of the projection of the unembedding vectors for $y \in \mathcal{Y}(w)$ should be small. To capture these requirements, we estimate the directions using a variant of Linear Discriminant Analysis (LDA), which finds a projection minimizing within-class variance and maximizing between-class variance. Formally, we estimate the vector representation of a binary feature $W$ for an attribute $w$ as

$$\bar{\ell}_w = \left(\tilde{g}_w^\top \mathbb{E}(g_w)\right) \tilde{g}_w, \quad \text{with} \quad \tilde{g}_w = \frac{\mathrm{Cov}(g_w)^\dagger \mathbb{E}(g_w)}{\|\mathrm{Cov}(g_w)^\dagger \mathbb{E}(g_w)\|_2}, \tag{5.2}$$

where $g_w$ is the unembedding vector of a word sampled uniformly from $\mathcal{Y}(w)$ and $\mathrm{Cov}(g_w)^\dagger$ is a pseudo-inverse of the covariance matrix. We estimate the covariance matrix $\mathrm{Cov}(g_w)$ using the Ledoit-Wolf shrinkage estimator (Ledoit & Wolf, 2004), because the dimension of the representation spaces is much higher than the number of samples.

### 5.2 WordNet Hierarchy is Linearly Represented

**Existence of Vector Representations for Binary Features** The first question is whether vector representations of binary features exist. To evaluate this, for each synset $w$ in WordNet we split $\mathcal{Y}(w)$ into train words (70%) and test words (30%), fit the LDA estimator to the train words, and examine the projection of the unembedding vectors for the test and random words onto the estimated vector

Table 1: Adding the linear representation of the parent concept to context embeddings does not affect the logit differences between token pairs in a child concept, and it substantially affects those in the parent concept. For each parent-child pair of binary concepts $(w_0 \Rightarrow w_1, z_0 \Rightarrow z_1)$, we first add the normalized linear representation $\bar{\ell}_W = \bar{\ell}_{w_1} - \bar{\ell}_{w_0}$ to the context embeddings. Then, we show the change in logit difference $\log \frac{\mathbb{P}(y_1 \mid \ell)}{\mathbb{P}(y_0 \mid \ell)}$ between the pairs $(y_0, y_1) \in \mathcal{Y}(w_0) \times \mathcal{Y}(w_1)$ (parent; top row) and $(y_0, y_1) \in \mathcal{Y}(z_0) \times \mathcal{Y}(z_1)$ (child; bottom row). Notice that, for any context $x$ and any tokens $y_0$ and $y_1$, adding a linear representation $\bar{\ell}_W$ for a binary contrast $W = w_0 \Rightarrow w_1$ manipulates the logit difference between the tokens from $\ell(x)^\top(g(y_1) - g(y_0))$ to $(\ell(x) + \bar{\ell}_W)^\top(g(y_1) - g(y_0))$, which implies the change in logit difference between the tokens is $\bar{\ell}_W^\top(g(y_1) - g(y_0))$, irrespective of $\ell(x)$. We show the mean and standard deviation of the change in logit differences over all token pairs.

| $W = \text{parent}_0 \Rightarrow \text{parent}_1$ & $Z = \text{child}_0 \Rightarrow \text{child}_1$ | Change in Logit Differences |
|---|---|
| $W =$ `plant.n.02` $\Rightarrow$ `animal.n.01` | $5.1265 \pm 1.1731$ |
| $Z =$ `mammal.n.01` $\Rightarrow$ `reptile.n.01` | $-0.0600 \pm 1.2190$ |
| $W =$ `fluid.n.02` $\Rightarrow$ `solid.n.01` | $9.8296 \pm 1.1099$ |
| $Z =$ `crystal.n.01` $\Rightarrow$ `food.n.02` | $0.3770 \pm 1.5410$ |
| $W =$ `scientist.n.01` $\Rightarrow$ `contestant.n.01` | $14.4222 \pm 0.9458$ |
| $Z =$ `athlete.n.01` $\Rightarrow$ `player.n.01` | $-0.1545 \pm 1.1426$ |

representation. The left plot in Figure 3 shows the mean and standard deviation of the projections, divided by the magnitude of each estimated $\bar{\ell}_w$. Following Theorem 4, if a vector representation exists for an concept, we would expect the values on the test set to be close to 1, and the values for random words to be close to 0. We see that this is indeed the case, giving evidence that vector representations do exist for these features.

As a baseline, the right plot in Figure 3 shows the same analysis but with the unembedding vectors randomly shuffled. In this case, the test projections are close to 0, which indicates that there are no linear representations. Thus, the existence of the linear representations in the original unembeddings relies on the underlying semantic structure.

**Intervention** Next, we validate that adding the estimated linear representations of a binary contrast changes the target concept without changing other off-target concepts, as required by Definition 3. Table 1 shows the mean and standard deviation of the changes in the logit differences between the pairs from parent or child binary contrasts, after adding the normalized linear representations of the parent binary contrast to context embeddings. The results show that the logit differences change significantly for the target concept, while the off-target concept changes very little.

**Relationship Between the Vector Representations of Binary Features** We now turn to examining whether the similarity between the vector representations of binary features reflects their semantic relation. The direct sum structure predicts that concepts that are close in semantic hierarchy should have similar vector representations. For example, `mammal` and `bird` are close in the hyponym graph because they share a common parent (`animal`). Our theory predicts that $\bar{\ell}_{\texttt{bird}}$ and $\bar{\ell}_{\texttt{mammal}}$ share a common component $\bar{\ell}_{\texttt{animal}}$ (the representation for "bird" is the representation for "animal" plus an orthogonal component, and similarly for mammal). This implies that the cosine similarity between $\bar{\ell}_{\texttt{mammal}}$ and $\bar{\ell}_{\texttt{bird}}$ should be substantial.

In Figure 4, we show the shortest distance between each feature in the (undirected) WordNet noun hyponym graph in the left panel and the cosine similarity between the estimated vector representations $\bar{\ell}_w$ in the middle panel. It is clear that, as predicted, the cosine similarity reflects the WordNet structure. As a control, we apply the same analysis after randomly shuffling the embeddings and show the resulting cosine similarities in the right panel. Here, direct parent-child (or grandparent-grandchild) relationships are still reflected in the cosine similarity because the shuffled embeddings still respect set inclusion (the words assigned to "mammal" are still a subset of those assigned to "animal"). However, sibling relationships are not reflected once the semantic structure is removed by the shuffling. In this case, the representations of siblings are effectively pairs of random vectors, and are nearly orthogonal as we would expect in a high dimensional space.

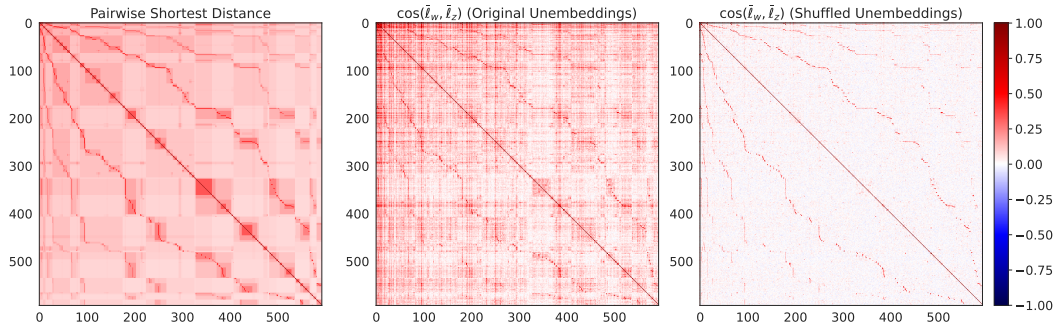


Figure 4: Hierarchical semantics in WordNet are linearly represented in Gemma-2B. The left heatmap shows pairwise shortest distance matrix between features in the noun hierarchy graph as $(1 + \text{min_distance})^{-1}$ (higher values indicate closeness, such as in child-parent or sibling relationships). The middle heatmap shows the cosine similarity between the vector representations $\bar{\ell}_w$. As predicted, this similarity reflects the WordNet structure. The right heatmap is a control where the embeddings are randomly shuffled (removing semantic structure). In this case, nearly everything is orthogonal, as expected in high-dimensional space (set inclusion relationships remain due to the estimation procedure). In Appendix F, we include zoomed-in versions of these heatmaps.

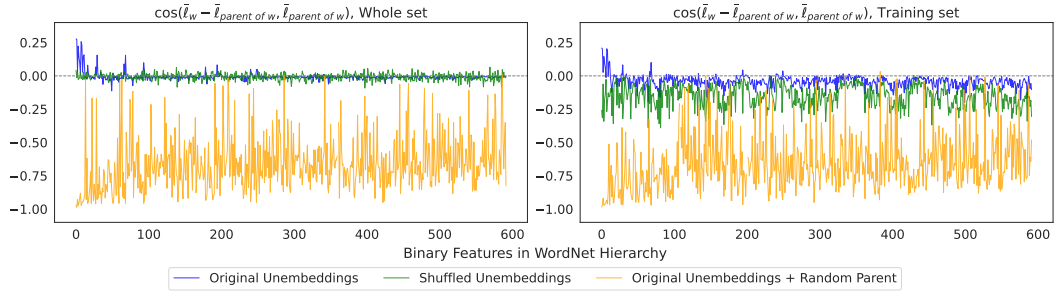


Figure 5: WordNet noun hierarchy is encoded in the orthogonal structure predicted by statement (a) in Theorem 8. We plot the cosine similarity between a child-parent vector and a parent vector for each feature in the hierarchy (blue). As predicted, this value is close to 0. The left plot uses all data for representation estimation, and the right plot uses only 70% independently selected for each synset. We include baselines where a randomly selected feature is used as the parent (orange) and where the embeddings are shuffled (green) as controls for the possibility that the orthogonality is a simple byproduct of high-dimensional geometry, or of the set inclusion relationships used in estimation—see main text for details. See Appendix F for an analogous plot for statement (d).

**Hierarchical Orthogonality** Finally, we evaluate the prediction that hierarchical relations are encoded orthogonally as predicted in Theorem 8. Figure 5 shows the cosine similarity $\bar{\ell}_{\texttt{parent}}$ and $\bar{\ell}_{\texttt{child}} - \bar{\ell}_{\texttt{parent}}$ for the WordNet features. As predicted, this value is close to 0 (blue curve).

Now, a challenge here is that in high dimensional spaces even random vectors are nearly orthogonal. So, it may be difficult to differentiate whether orthogonality reflects semantic structure or is merely a consequence of nearly everything being orthogonal in high dimensions. As a control, we also show the cosine similarity when the parent vector is the representation of a randomly selected feature (orange). In this case, the cosine similarity is far from 0, suggesting that the observed orthogonality is not merely a byproduct of the high-dimensional geometry.

We also include another baseline that estimates the cosine similarity between the child-parent vector and the parent vector for each feature using shuffled unembeddings (green). In the left plot, we see that we still have orthogonality, which could suggest that the orthogonality is a consequence of the set inclusion relations in WordNet (rather than actual semantic structure).[5] To test this, in the right plot we estimate the representations using only 70% of the tokens, independently selected for each

[5]In Appendix G, we explain why set inclusion (before train/test split) leads to orthogonality.